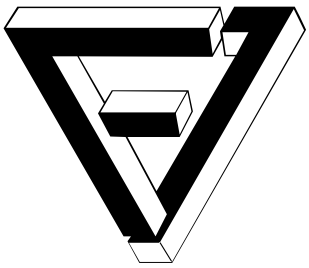
Masaryk University
Faculty of Informatics



Side-Effect Analysis Using a Static Single Use Program Representation

Master's Thesis

Samuel Malec

Brno, 2026

Declaration

I hereby declare that this thesis is my original work and that I developed it independently. I have properly cited all sources and literature used in this thesis and provided complete references to them.

Supervisor: TBD

Acknowledgements

Write your acknowledgements here.

Abstract

Write your abstract here.

Keywords

keyword, keyword

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I Introduction

2 Preliminaries

In this chapter we will introduce the necessary terms that underpin the core ideas presented in this thesis. The industrial revolution and it's consequences have been a disaster for the human race. Program analysis is used to infer facts about program behavior of interest. In the current events such as the emergence of LLM which significantly increased the pace of software development, it is more than ever necessary to analyse programs to reason about their correctness. Although program analysis has been a subject of active research for well over half a century, there haven't been many significant ground breaking innovations in the last few years. …

2.1 Side Effect Analysis

2.1.1 Mod/Ref Summaries

2.1.2 Alias and Points-to Analysis

2.2 Static Single Use Form

2.2.1 SSA as Functional Programming

2.2.2 Linear Values

2.2.3 Cthu

2.3 Sea of Noddes

2.4 GraalVM and Graal IR

2.4.1 Fixed and Floating Nodes

2.4.2 The Memory Graph and Location Identities

2.4.3 Calls, Exceptions and Deoptimizations

2.4.4 Memory Optimizations in Graal

2.5 Native Image

2.5.1 The Closed World

2.5.2 Points-to Analysis

2.5.3 Context Sensitivity

3 Analysis and Optimization of the Explict Effect Form

4 From Graal IR and back

I went to hell and back just to find peace

5 Evaluation

6 Conclusion

Bibliography